35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — MILES (1 mile = 63,360 inches)—continued

FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION Scale Unit :	represents Map Unit	TO FIND MILES PER INCH (**x** in ratio) Use ratio below or $\frac{SCALE}{63\,360}$	MILES PER INCH	TOTAL MILES ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (**y** in ratio) Use ratio below or $\frac{\text{Miles on scale}}{\text{Miles per inch}}$	TOTAL SCALE LENGTH (INCHES)
1:250 000	1 inch	250 000 in	$\frac{63\,360}{1} = \frac{250\,000}{\mathbf{x}}$	3.945707	25	$\frac{3.945707}{1} = \frac{25}{\mathbf{y}}$	6.336
1:400 000	1 inch	400 000 in	$\frac{63\,360}{1} = \frac{400\,000}{\mathbf{x}}$	6.3131313	40	$\frac{6.3131313}{1} = \frac{40}{\mathbf{y}}$	6.336
1:500 000	1 inch	500 000 in	$\frac{63\,360}{1} = \frac{500\,000}{\mathbf{x}}$	7.8914141	50	$\frac{7.8914141}{1} = \frac{50}{\mathbf{y}}$	6.336
1:750 000	1 inch	750 000 in	$\frac{63\,360}{1} = \frac{750\,000}{\mathbf{x}}$	11.837121	60	$\frac{11.837121}{1} = \frac{60}{\mathbf{y}}$	5.068
1:1 000 000	1 inch	1 000 000 in	$\frac{63\,360}{1} = \frac{1\,000\,000}{\mathbf{x}}$	15.782828	125	$\frac{15.782828}{1} = \frac{125}{\mathbf{y}}$	7.920
1:2 000 000	1 inch	2 000 000 in	$\frac{63\,360}{1} = \frac{2\,000\,000}{\mathbf{x}}$	31.565656	250	$\frac{31.565656}{1} = \frac{250}{\mathbf{y}}$	7.920
1:2 500 000	1 inch	2 500 000 in	$\frac{63\,360}{1} = \frac{2\,500\,000}{\mathbf{x}}$	39.45707	300	$\frac{39.45707}{1} = \frac{300}{\mathbf{y}}$	7.603
1:5 000 000	1 inch	5 000 000 in	$\frac{63\,360}{1} = \frac{5\,000\,000}{\mathbf{x}}$	78.914141	600	$\frac{78.914141}{1} = \frac{600}{\mathbf{y}}$	7.603
1:7 500 000	1 inch	7 500 000 in	$\frac{63\,360}{1} = \frac{7\,500\,000}{\mathbf{x}}$	118.37121	600	$\frac{118.37121}{1} = \frac{600}{\mathbf{y}}$	5.068
1:10 000 000	1 inch	10 000 000 in	$\frac{63\,360}{1} = \frac{10\,000\,000}{\mathbf{x}}$	157.82828	1100	$\frac{157.82828}{1} = \frac{1100}{\mathbf{y}}$	6.969

To find miles per inch on 1: 250 000 map . . .

63,360 inches = 1 mile
Show in ratio as ...

$$\frac{63\,360}{1}\ \frac{\textit{inches}}{\textit{miles}}$$

Let SCALE (250 000) be in inches
Fractional scale says 1 inch represents 250,000 in
Let **x** be miles that 1 inch represents on map
Show in ratio as ...

$$\frac{250\,000}{\mathbf{x}}\ \frac{\textit{inches}}{\textit{miles}}$$

Solution . . .

$$63\,360 \cdot \mathbf{x} = 250\,000 \cdot 1$$

$$\frac{63\,360}{1} = \frac{250\,000}{\mathbf{x}}$$

$$\frac{63\,360\,\mathbf{x}}{63\,360} = \frac{250\,000}{63\,360}$$

$$\mathbf{x} = \frac{250\,000\ \text{(SCALE)}}{63\,360}$$

$$\mathbf{x} = 3.945707$$